GOKULAN S

Embedded Systems Engineer | Firmware Developer

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PROFILE

Embedded Systems Engineer targeting Embedded Software, Firmware, Embedded Systems, and Automotive Embedded roles, with hands-on experience in Embedded C, C, STM32/ARM Cortex-M, PIC, bare-metal firmware, peripheral drivers, hardware bring-up, PCB analysis, and debugging. Skilled in GPIO, ADC, PWM, timers, interrupts, UART, I2C, SPI, EEPROM, RTC, and sensor integration. Experienced in real-time monitoring, functional testing, motor diagnostics, and hardware-software integration, including identification of 3 critical BLDC motor winding faults contributing to a 10% power-consumption reduction

CORE SKILLS

- **Programming:** Embedded C, C, Python
- **MCUs & Architectures:** STM32, ARM Cortex-M, PIC16F877A, PIC18F23K22, LPC1768 Cortex-M3, ESP8266, ESP32
- **Embedded Systems:** Bare-Metal Firmware, Register-Level Programming, Peripheral Drivers, GPIO, ADC, PWM, Timers/Counters, Interrupts, Real-Time Data Acquisition
- **Communication:** UART, I2C, SPI, MQTT
- **Peripherals:** EEPROM, RTC, HD44780 LCD, IR Speed Sensor, Temperature & Vibration Sensors
- **Hardware & Debugging:** Hardware Bring-Up, PCB Analysis, HW/SW Debugging, Sensor Integration, Functional Testing, Fault Detection, Root Cause Analysis
- **Tools:** MPLAB X IDE, XC8 Compiler, Proteus 8, KiCad, Git, GitHub

PROFESSIONAL EXPERIENCE

INNOVENDUS TECHNOLOGY PRIVATE LIMITED **March 2026 – Present**

Embedded Systems Engineer

- Contribute to the embedded product development lifecycle, covering hardware bring-up, bare-metal firmware development, validation, debugging, and functional testing.
- Develop Embedded C firmware and register-level peripheral drivers for microcontroller-based products.
- Perform PCB analysis and hardware bring-up, validating hardware performance and supporting reliable hardware-software integration.
- Troubleshoot embedded hardware and firmware issues during development and functional validation.

WATTIC SWAMINATHAN ENTERPRISES PVT LTD **October 2025 – March 2026**

Graduate Engineer Trainee – Motor Testing & R&D

- Performed Motor Comprehensive Tests (MCT) and no-load testing on BLDC motors used in AC indoor and outdoor units.
- Identified 3 critical winding faults in supplier PCB BLDC motors through systematic testing and technical analysis.
- Contributed to a 10% reduction in power consumption through identification and resolution of motor-related faults.
- Conducted root cause analysis, dynamometer testing, and top-load testing on induction and washing-machine motors.

EDUCATION

SRI MANAKULA VINAYAGAR ENGINEERING COLLEGE **2021 – 2025**

B.Tech – Electronics and Communication Engineering

PROJECTS

INDUSTRIAL MOTOR CONDITION MONITORING & FAULT LOGGING SYSTEM

PIC16F877A | Embedded C | MPLAB X | XC8 | UART | I2C | ADC | EEPROM

- Developed a bare-metal Embedded C firmware system for real-time industrial motor health monitoring and fault logging..
- Integrated LM35 temperature and vibration sensors through ADC for continuous condition monitoring.

BATTERY HEALTH MONITORING & DATA LOGGING SYSTEM

PIC16F877A | Embedded C | ADC | I2C | EEPROM | LCD

- Developed firmware for ADC-based voltage and current sensing with real-time State of Charge (SoC) estimation.
- Implemented an I2C EEPROM driver to store timestamped battery parameters for historical monitoring and analysis.

CERVICAL CANCER DETECTION USING DEEP LEARNING

CNN | TensorFlow | Keras | OpenCV | Scikit-learn | IEEE Conference Paper

- Trained a Convolutional Neural Network (CNN) using 906 colposcopy images from 200 patients for CIN-stage classification.
- Achieved approximately 85% validation accuracy in cervical intraepithelial neoplasia stage detection.

CERTIFICATIONS & TRAINING

Embedded Systems – 6-Month Industry Training — April 2026 – Present

C Programming – 40 Hours — 2022

ADDITIONAL INFORMATION

Github: https://github.com/gokulan-s
Portfolio: https://sgokulan.netlify.app/

LANGUAGES

English, Tamil, French